Grant Yang

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Skills

Languages: Lean4, Python, C/C++, Rust, Java, TypeScript.

Technology: PyTorch, OpenGL, Git, Linux/Unix, Networking.

Experience

UW Math AI Lab - Student Researcher (Spring 2026 - current) monogenic-exetensions

- Worked with Prof. Bianca Viray to formalize select results from her paper [2503.07846] using Lean. Namely, that finite étale extensions of local rings are monogenic.
- Contributed code to Mathlib: PR#36421 and PR#37527.

Euler Circle - Teaching Assistant (Summer 2026) Mathematics in Lean

- Taught class on real analysis and Lean for advanced high school students.
- Held office hours/homework sessions and graded student work.

Personal Projects

UW Lean Hackathon - 1st Place leomayer1/cat-rw, uw2026leanhackathon.github.io

- Won 1st prize for a tactic to peform rewriting using isomorphisms rather than equalities.
- Automated the process of writing tedious proofs, especially useful for category theory.

ByteGPT Online demo

- Trained a tiny 20M-parameter transformer using PyTorch with a custom architecture.
- Hand-wrote a highly efficient C++ inference engine targeting WebAssembly.
  - Runs in-browser at FP32 using only 300 MB for 4096 context, at over 50 tok/s.

C++ Chess Engine ROTARTSI82/chess

- Wrote a multithreaded chess engine with magic bitboards and alpha-beta search.
- Optimized to search millions of positions per second, reaching over 1600 Lichess rating.

Pascal Compiler ROTARTSI82/ATCS

- Designed and built a compiler for a custom Pascal dialect targeting MIPS32 assembly.
- Demonstrated complex programs on a MIPS32 emulator.

4-bit Computer Blog post

- Using KiCAD, I designed and built a working 4-bit computer on a custom PCB, starting from 7400-series TTL chips.
- Demonstrated simple programs like computing the Fibonacci numbers.

Education

University of Washington Seattle, WA

B.S. in Electrical and Computer Engineering *2025 – 2029*

- **Coursework**: MATH 208: Linear Algebra, MATH 327: Real Analysis, MATH 380: Computational Algebraic Geometry, STAT 390: Intro to Statistics.
- **GPA**: *3.99/4.0*

Harker Upper School San Jose, CA

4.6/4.8 GPA *2021 – 2025*